

ARAFAT HOSSAIN FAHIM

AI/NLP Engineer | Python/Django Developer | Full-Stack Intelligence Systems Builder

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[GitHub](#) | [Hugging Face](#) | [LinkedIn](#) | Portfolio: [Add portfolio URL]

PROFESSIONAL SUMMARY

AI/NLP engineer and full-stack systems builder specializing in Bangla NLP, transformer fine-tuning, Django/Node.js backends, data pipelines, social/news intelligence platforms and automation products. Experienced in building end-to-end systems that collect noisy web, news, social and media data; normalize it; apply ML/NLP analysis; and expose it through dashboards, APIs and operational workflows. Published Bangla NLP models and datasets for sentiment, emotion, NER, news classification and event extraction.

CORE TECHNICAL SKILLS

AI / NLP: Transformers, Hugging Face, PyTorch, BanglaBERT/ELECTRA, text classification, token classification, NER, sentiment/emotion analysis, event extraction, model evaluation

Backend / APIs: Python, Django, Django REST Framework, Flask, Node.js, Express, ASGI/Daphne, authentication, RBAC, REST APIs, admin dashboards

Data / Databases: PostgreSQL, SQLite, MongoDB, Redis, pandas, NumPy, scikit-learn, joblib, structured datasets, ETL/data normalization

Frontend / Product: React, Vite, TypeScript basics, Tailwind CSS, Bootstrap, Chart.js, Recharts, dashboard UI, real-time progress and analytics views

Automation / DevOps: Docker, Docker Compose, Selenium, Playwright, Telethon, ffmpeg, OpenCV, WebSockets, background workers/schedulers, Git/GitHub

PROJECT-BASED EXPERIENCE

Independent AI/NLP Engineer and Full-Stack Systems Builder | Dhaka, Bangladesh / Remote

- Built production-style intelligence platforms for election monitoring, social media monitoring, news scraping, event extraction, trending-topic analysis and structured decision support.
- Designed multi-service systems with web dashboards, REST APIs, background schedulers, scraper workers, Docker deployments and database-backed operational workflows.
- Fine-tuned and published Bangla NLP models for sentiment/emotion, named entity recognition, news topic classification and crime-event detection use cases.
- Developed automation products spanning WhatsApp campaigns, AI auto-replies, video-to-reel extraction, PostgreSQL management, image hosting and workflow/task APIs.

PUBLISHED AI MODELS AND DATASET

Sentimotion - Joint Bangla Sentiment and Emotion Model - BanglaBERT Large, PyTorch, Hugging Face Transformers

- Built a multi-task Bangla NLP model with shared encoder and separate prediction heads for sentiment and emotion classification.
- Positioned for Bangla social media analytics, political discourse monitoring, opinion mining and moderation-support workflows.

BanglaTag - Bengali Named Entity Recognition Model and Dataset - BanglaBERT/ELECTRA, BIO tagging, Hugging Face Datasets

- Fine-tuned a Bengali NER model and published an aligned token/tag dataset for entities such as persons, locations, organizations, political entities, events and crimes.
- Prepared reusable NER assets for downstream information extraction, news intelligence and knowledge-graph pipelines.

Crime Event Detection and Bangla News Topic Classification - BanglaBERT, multi-task classification/extraction

- Developed Bangla news models for topic classification and structured crime-event extraction, including event type, location/time spans and article-level classification outputs.
- Supported the intelligence-system direction by converting unstructured Bangla news articles into structured labels and entities.

SELECTED SYSTEMS PROJECTS

Prism - Election and News Intelligence Platform - Django 5, PostgreSQL, Daphne/ASGI, APScheduler, ML/NLP, Docker

- Built a dashboard-driven intelligence platform for election reports, geography, candidates, officials, seat-wise news, trending topics, event extraction, ML event detection and sentiment analysis.
- Implemented modules for reports, evidence, candidate/party/official management, geographic hierarchy, live results, trending-topic analysis and Telegram delivery support.

SocialInt - Social Media Intelligence and Monitoring Platform - Django 5.2, DRF, PostgreSQL, Daphne, Docker

- Built a social monitoring platform that stores normalized posts, authors, media, source configurations, scraping credentials and keyword-analysis results.
- Implemented dashboard and API workflows for Facebook, TikTok, YouTube and configurable social-source monitoring with role-based access.

All in One Scraper - Distributed News and Social Scraper Suite - Python, Docker Compose, Selenium, Playwright, Telethon, twikit, BeautifulSoup

- Created a multi-service scraper system for Bangladeshi news portals, Facebook, X/Twitter, YouTube, TikTok, Telegram and Google News sources.
- Designed each scraper as an independent Dockerized service that collects recent posts/articles, normalizes output and sends data to a backend API.

Reel Maker - AI-Powered Short-Form Video Generator - Django, DRF, Channels, React, Redis, Huey, OpenCV, ffmpeg

- Built a tool that analyzes long-form videos, scores hook moments using audio energy, visual motion/change and subtitle patterns, then exports vertical social-media reels.
- Implemented upload, chunking, background analysis, WebSocket progress updates, clip streaming/download and customizable text overlays.

WhatsApp Automation System - Node.js, Express, React, TypeScript, MongoDB, Redis, Socket.io, OpenAI API

- Built a full-stack WhatsApp automation platform with QR login, AI-powered auto-replies, campaign scheduling, contact management, media sending and analytics.
- Used JWT authentication, Redis session storage, real-time Socket.io updates and OpenAI-powered response generation.

MultiBase - PostgreSQL Database Management System - Django, PostgreSQL, Bootstrap 5, Chart.js, Docker Compose

- Developed a web-based PostgreSQL management console for creating/managing databases, users, connection strings, table/size views and database metrics.
- Included monitoring charts and a Docker setup with Django web UI plus managed PostgreSQL server connectivity.

Workflow API - Role-Based Task and Team Management Backend - Node.js, Express, SQLite, JWT, file storage

- Built a backend MVP for authentication, users, teams, task assignment, confidential tasks, submissions, comments, attachments, notifications, reports and audit logs.
- Implemented role-based access for System Admin, OIC, Teamleader and Users, with task review/reopen and overdue-task workflows.

ImageStash - Dockerized Image Hosting App - Flask, Docker, Docker Compose

- Built a lightweight image hosting and management application with local development and Docker deployment workflows.

EDUCATION

B.Sc. in Computer Science and Engineering | January 2018 - January 2022

IUBAT - International University of Business Agriculture and Technology

CGPA: 3.40/4.00

Higher Secondary Certificate (HSC) | June 2015 - June 2017

Dhaka College, Dhaka

Group: Science | GPA: 5.00/5.00

Secondary School Certificate (SSC) | January 2010 - June 2015

Motijheel Govt. Boys' High School

Group: Science | GPA: 5.00/5.00

OPEN SOURCE, ROLES AND ADDITIONAL INFORMATION

Portfolio: GitHub shows Python/Django, ML, Node.js, React, automation, APIs, scrapers, system utilities and full-stack projects; Hugging Face shows Bangla NLP models/datasets.

Best-fit roles: AI/NLP Engineer, Python/Django Developer, Machine Learning Engineer, Data/Intelligence Systems Builder, Full-Stack Developer.

Languages: Bengali/Bangla, English

Location: Dhaka, Bangladesh - open to remote, hybrid or onsite roles [edit as needed]

Availability: [Add notice period / availability]